How long should a block be?

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(joint work with Anthony Davison)

Block maximum method

Idea promoted for engineering applications by Emil Gumbel (cf. Gumbel 1958).

- divide a (time) series of length n into n_m equal-length disjoint blocks of m successive observations,
- fit the generalized extreme value (GEV) distribution to the resulting sample of n_m block maxima,
- use the fitted model for risk assessment.

This talk: how long should a block be?

Motivating example

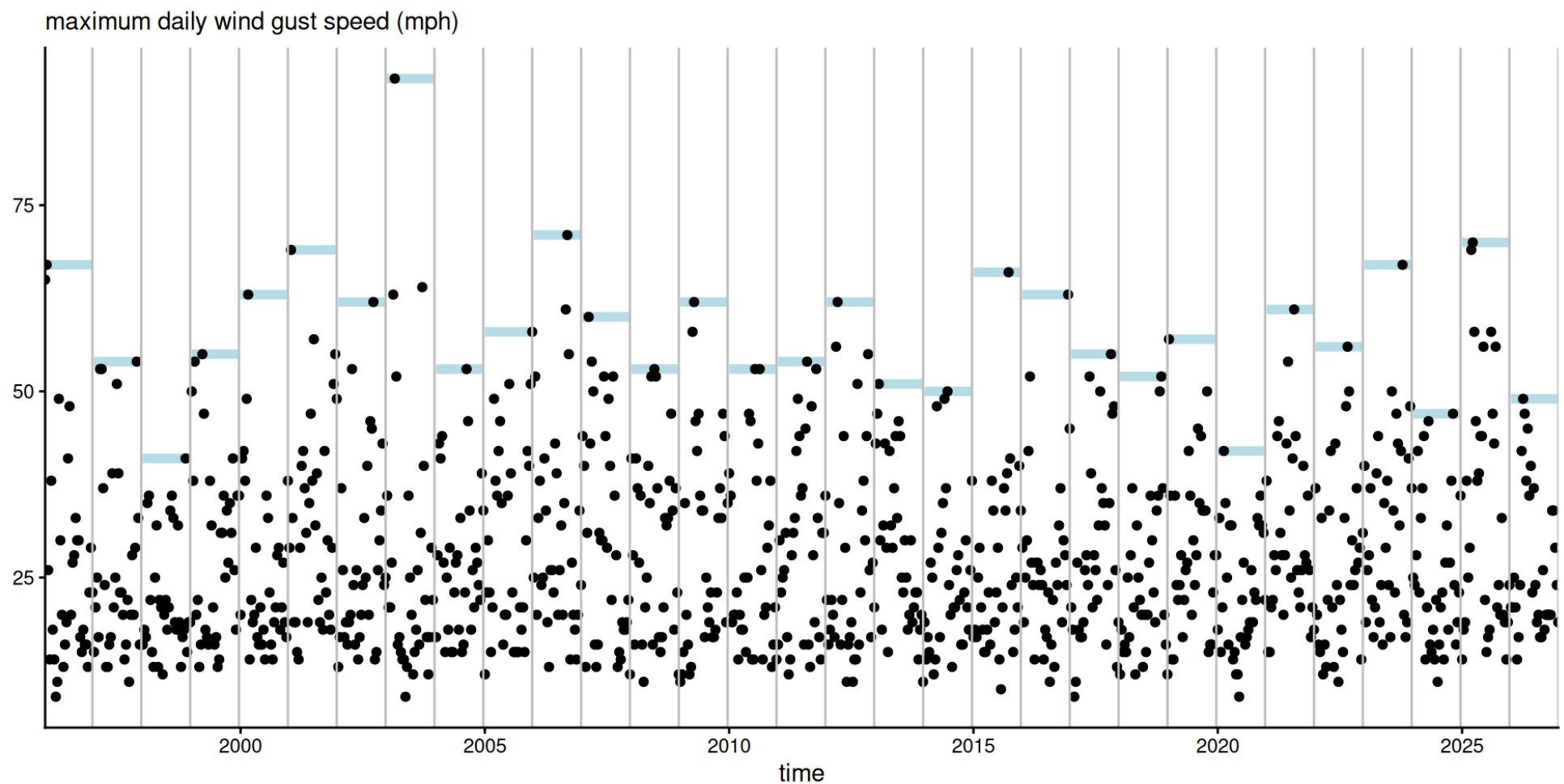


Figure 1: Daily maximum wind gust speed in Cheeseboro, California, for January 1st to February 1st of 1996–2026, for a total of 31×32 measurements.

Extremal types theorem

The justification for using the GEV comes from Fisher and Tippett (1928), Mises (1936), Gnedenko (1943).

If G is a distribution function and there exist location and scaling sequences d_m and $c_m > 0$ such that the maximum of m observations has a non-degenerate limiting distribution

$$\lim_{m \to \infty} G^m(c_m x + d_m) = F(x)$$

then $F(x)$ must be a GEV distribution function.

Generalized extreme value distribution

The GEV distribution function is

$$F(x;\mu,\sigma,\xi) = \begin{cases} \exp\left[-\left\{1+\xi\frac{x-\mu}{\sigma}\right\}_+^{-1/\xi}\right], & \xi \neq 0, \\ \exp\left[-\exp\left\{-\frac{x-\mu}{\sigma}\right\}\right], & \xi = 0. \end{cases}$$

with location $\mu \in \mathbb{R}$, scale $\sigma > 0$ and shape $\xi \in \mathbb{R}$.

The support of $\mathsf{GEV}(\mu,\sigma,\xi)$ is $\{x : \sigma + \xi(x-\mu) > 0\}$.

Regularity conditions and normalizing constants

Suppose that $\lim_{m\to\infty} G^m(c_m x + d_m) = F(x)$ for $F(x) \sim \text{GEV}(0, 1, \xi)$.

- Assume G is twice differentiable with possibly infinite endpoint $x^\star$, with density g and define $\phi(x) = -G(x)\log G(x)/g(x)$.
- for convergence to the GEV distribution, it is sufficient that $\lim_{x\to x^\star} \phi'(x) = \xi^\star \in \mathbb{R}$ (so-called von Mises condition).
- The normalizing constants are obtained from $-\log G(d_m) = m^{-1}$ and $c_m = \phi(d_m)$. The speed of convergence depends on the (unknown) G.
- Penultimate approximation (Smith 1987) is $\text{GEV}\{d_m, c_m, \phi'(d_m)\}$.

Quality of approximations

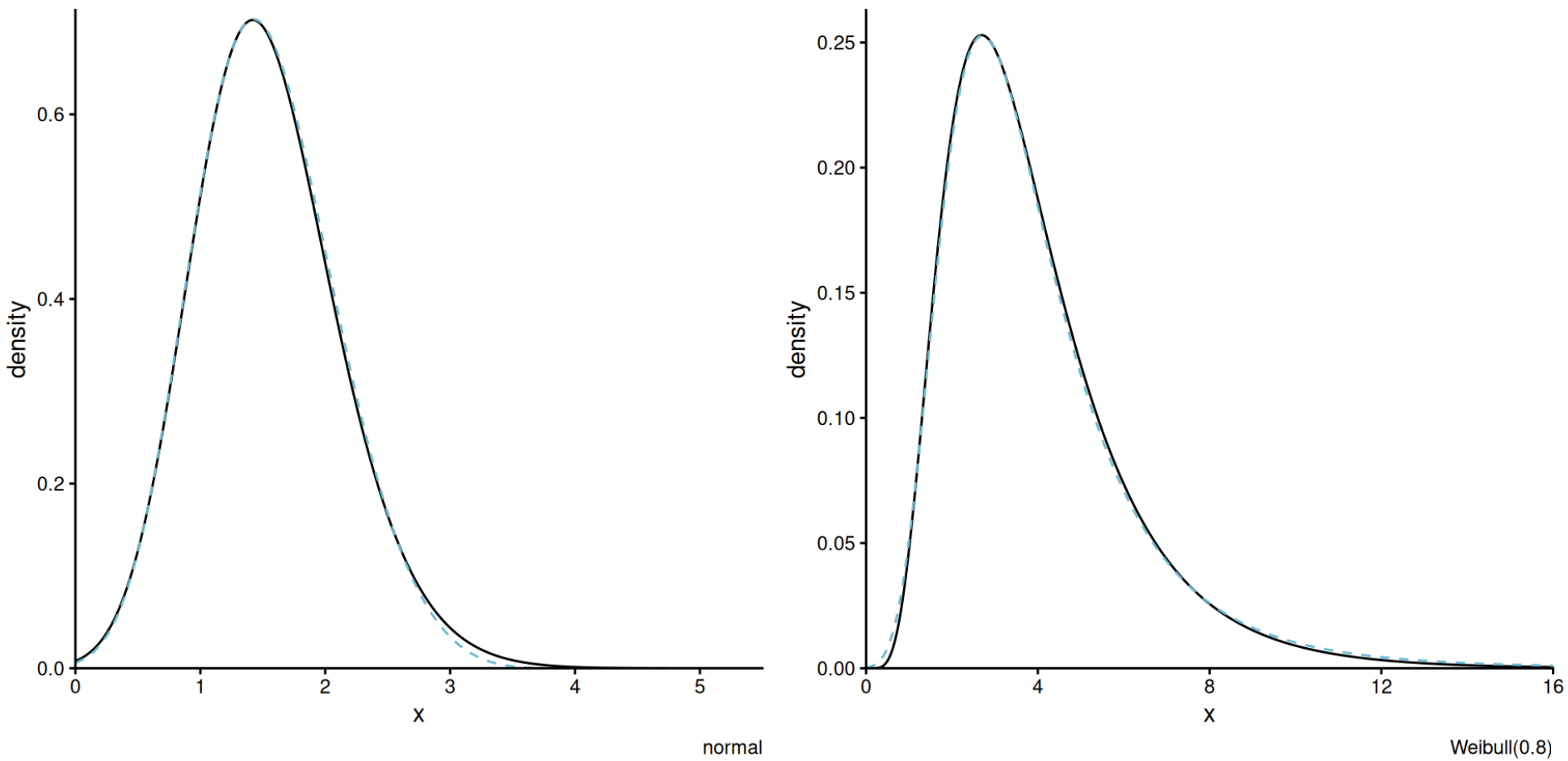


Figure 2: Densities of GEV penultimate approximations (dashed blue) versus true distribution G^{30} (full black).

Max-stability

The GEV is characterized by *max-stability*:

if $X_j \overset{\text{iid}}{\sim}$ GEV(μ, σ, ξ), then for any m,

$$Y = \max\{X_1, \ldots, X_m\} \sim \text{GEV}(\mu_m, \sigma_m, \xi),$$

with

$$\begin{aligned}
\mu_m &= \mu + \sigma(m^\xi - 1)/\xi, & \sigma_m &= \sigma m^\xi, & &\text{if } \xi \neq 0,\\
\mu_m &= \mu + \sigma \log m, & \sigma_m &= \sigma, & &\text{if } \xi = 0.
\end{aligned}$$

Penultimate approximation vs max-stability

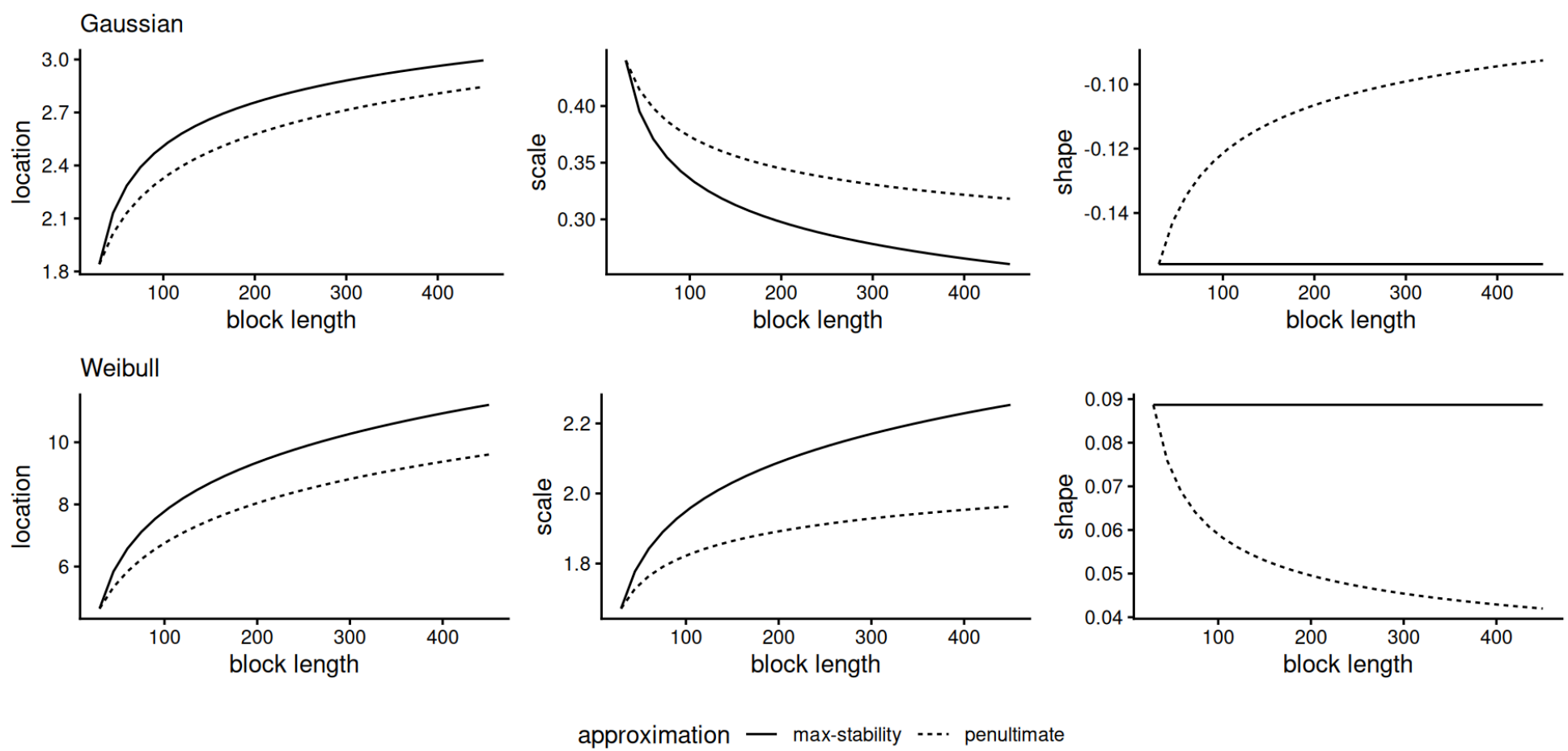


Figure 3: GEV parameters based on max-stable (dashed), and the penultimate approximation (full) for Gaussian (top) and Weibull (bottom).

Research question: How to choose the block length?

Traditional extreme value bias-variance trade-off:

- Smaller block lengths may lead to poor GEV approximation (potentially biased extrapolation).
- Longer blocks lead to smaller sample sizes (more variability).

Other considerations

- Data availability (frequency of records).
- Block maximum is widely viewed as the more robust approach in the applied literature (compared to peaks-over-threshold).
- Theoretical consideration of asymptotic efficiency relative to peaks-over-threshold (Bücher and Zhou 2021).
- Natural (minimum) block length (e.g., annual for seasonally-varying data).

Literature review

Surprisingly, very little work in the area.

- Wang et al. (2016) suggest using combinations of goodness-of-fit tests (such as Anderson–Darling or Kolmogorov–Smirnov).
- unpublished work of Dkengne et al. (2020) suggests parameter stability plot built using max-stability.

Testing for different block lengths

Consider nm block maxima for $n, m \in \mathbb{N}$. We gather observations into an $n \times m$ matrix of ordered inputs.

Table 1: Blocks of size $m = 4$ from daily maxima wind gusts (in km/h).

14	26	65	67
9	14	18	38
11	20	30	49
13	16	19	41
20	27	28	48

Row i contains the order statistics $X_{i,(1)} \leq \cdots \leq X_{i,(m)}$.

Joint distribution of order statistics

Let $X_{(1)} \leq \cdots \leq X_{(m)}$ denote order statistics.

Let F and f be the CDF and density of $X_j \overset{\text{iid}}{\sim} \text{GEV}(\mu, \sigma, \xi)$.

The joint density of order statistics is the product of

- the conditional $X_{(1)}, \ldots, X_{(m-1)} \mid X_{(m)} = x_{(m)}$,
- the marginal $X_{(m)}$.

Conditional density of lower order statistics

The conditional density of $X_{(1)}, \ldots, X_{(m-1)} \mid X_{(m)}$ is

$$f_{(1,\ldots,m-1)|(m)}(\boldsymbol{x}_{-(m)}) = (m-1)! \prod_{j=1}^{m-1} \frac{f(x_{(j)}; \mu, \sigma, \xi)}{F(x_{(m)}; \mu, \sigma, \xi)}$$

so the $(m-1)$ smallest observations are, up to permutation, $\mathsf{GEV}(\mu, \sigma, \xi)$ truncated above at $x_{(m)}$.

Distribution of block maximum

The block maximum $Y = \max(X_1, \ldots, X_m)$ has CDF $F_{(m)}(x) = F^m(x)$ and density

$$\begin{aligned} f_{(m)}(x) &= mF(x; \mu, \sigma, \xi)^{m-1} f(x; \mu, \sigma, \xi) \\ &= f(x; \mu_m, \sigma_m, \xi), \end{aligned}$$

by max-stability.

Likelihood

In the simplest case, we have

$$\mathcal{L}(\boldsymbol{\vartheta}_0, \boldsymbol{\vartheta}_m) = \prod_{i=1}^{n} f_{(1,\ldots,m-1)|(m)}(\boldsymbol{x}_{i,-(m)}; \boldsymbol{\vartheta}_0) f_{(m)}(x_{i,(m)}; \boldsymbol{\vartheta}_m),$$

where

- $\boldsymbol{\vartheta}_0 = (\mu, \sigma, \xi)$ are the parameters for $X_{i,(1)} \leq \cdots \leq X_{i,(m-1)}$.
- $\boldsymbol{\vartheta}_m$ are the parameters of the largest observation $X_{i,(m)}$.

Under the null hypothesis of max-stability,
$\boldsymbol{\vartheta}_m = (\mu_m, \sigma_m, \xi) = g(\boldsymbol{\vartheta}_0)$, say.

Rounding and left-censoring

Environmental measurements are frequently rounded to the nearest δ unit. We may also want to left-censor observations below u.

Consider latent $X_i^* \sim \text{GEV}(\mu, \sigma, \xi)$, where $X_{i,j} - \delta/2 \leq X_i^* \leq X_{i,j} + \delta/2$; the individual observation contribution to the likelihood when $x_{i,(j)} > u$ are

$$L_{i,j}(\boldsymbol{\vartheta}) = \begin{cases} f(x_{i,(j)}; \boldsymbol{\vartheta}), & \delta = 0; \\ F(x_{i,(j)} + \delta/2; \boldsymbol{\vartheta}) - F(\max\{x_{i,(j)} - \delta/2, u\}; \boldsymbol{\vartheta}), & \delta > 0. \end{cases}$$

The joint likelihood contributions for the $(m-1)$ lowest order statistics of m are from the GEV distribution right-truncated at $x_{(m)} + \delta/2$.

Joint likelihood with censoring

The likelihood contribution from a single vector $x_{i,(1)}, \ldots, x_{i,(m)}$ is

$$L_i(\boldsymbol{\vartheta}_0, \boldsymbol{\vartheta}_m) = L_{i,m}(\boldsymbol{\vartheta}_m)^{w_{i,(m)}(\boldsymbol{\vartheta}_m)} F(u; \boldsymbol{\vartheta}_m)^{1-w_{i,(m)}(\boldsymbol{\vartheta}_m)} \times \prod_{j=1}^{m-1} \frac{L_{i,j}(\boldsymbol{\vartheta}_0)^{w_{i,(j)}(\boldsymbol{\vartheta}_0)} F(u; \boldsymbol{\vartheta}_0)^{1-w_{i,(j)}(\boldsymbol{\vartheta}_0)}}{F\left(\max\{u, x_{i,(m)} + \delta/2\}; \boldsymbol{\vartheta}_0\right)}$$

where the weight $w_{i,j}(\boldsymbol{\vartheta})$ indicates the probability that $X^*_{i,j} > u$ (expected likelihood) (Section 2.2 of Varty et al. 2021).

Alternative hypotheses

The idea underlying our testing procedure is to let the parameters of the block maximum differ, by

- A_1: setting $\mathcal{H}_a : \boldsymbol{\vartheta}_m = (\mu_{\omega m}, \sigma_{\omega m}, \xi)$ for some scaling $\omega > 0$, and testing $\mathcal{H}_0 : \omega = 1$.
- A_2: allowing both location and scale to vary $\mathcal{H}_a : \boldsymbol{\vartheta}_m = (\mu_m + \nu, \phi\sigma_m, \xi)$ and testing $\mathcal{H}_0 : \nu = 0, \phi = 1$.
- A_3: letting all three parameters vary $\mathcal{H}_a : \boldsymbol{\vartheta}_m = (\mu_m + \nu, \phi\sigma_m, \xi + \zeta)$ and testing $\mathcal{H}_0 : \nu = 0, \phi = 1, \zeta = 0$.

We use likelihood ratio statistics to compare the models $\mathcal{H}_0$ and $\mathcal{H}_a$. The asymptotic distribution of the test is χ^2_k with $k \in \{1, 2, 3\}$.

Setup of power study

We consider different scenarios to assess the power of tests:

- different parametric distributions for alternatives,
- sample sizes $n \in \{25, 50, 100\}$,
- number of order statistics $m \in \{2, 5, 10\}$,
- three alternatives $\mathcal{H}_a$.

All power curves are based on 2000 replications, with tests conducted at level 5%.

Alternative distributions

We consider maxima of

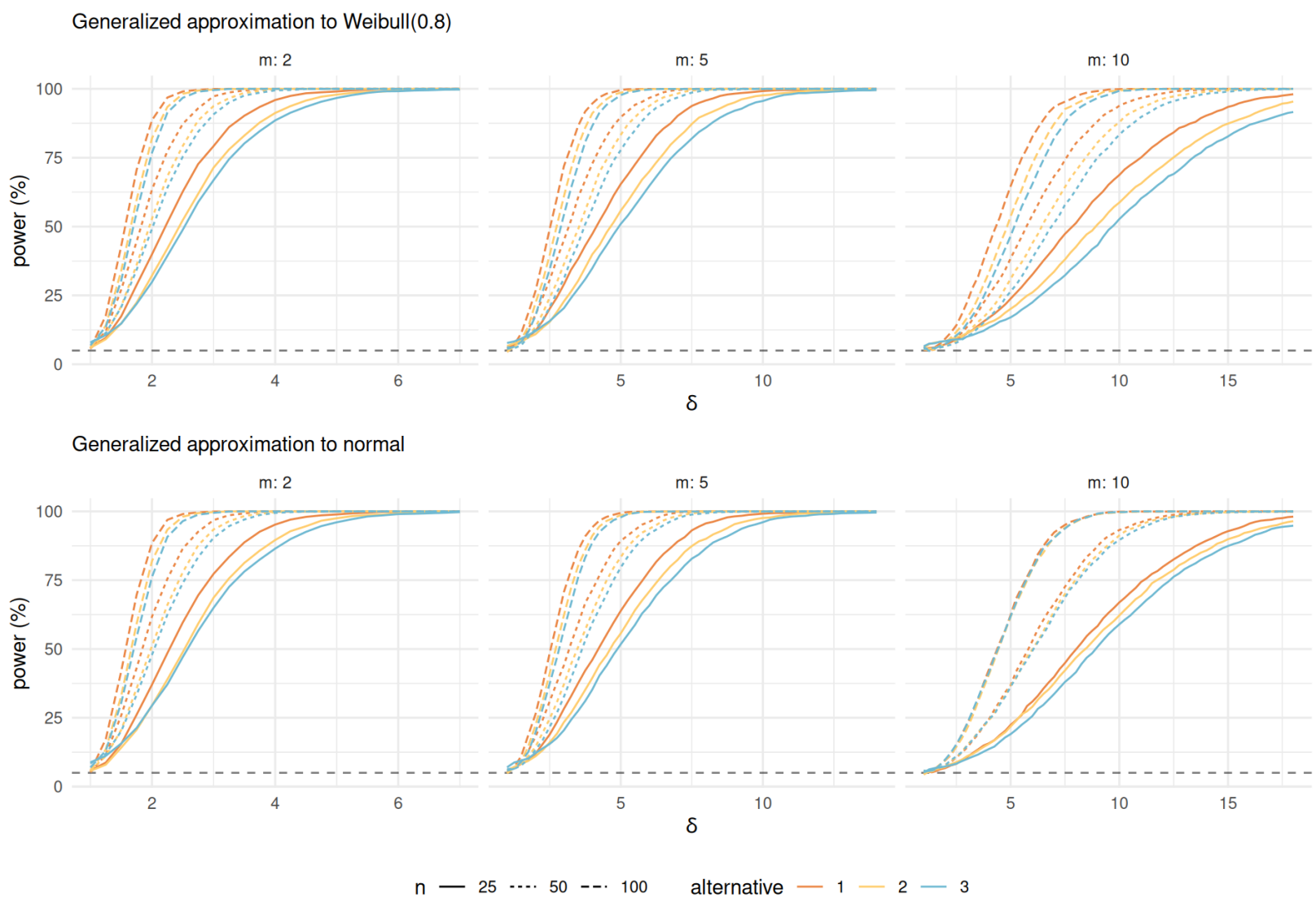
a. **normal**$(0, 1)$ and

b. **Weibull**$(1, 0.8)$.

with distribution G and block of size 30, simulate

- $X_{(1)}, \ldots, X_{(m-1)}$ from G^{30},
- $X_{(m)}$ from $G^{30\delta}$, but left-truncated at $x_{(m-1)}$.

Power curves



Size of tests

		m	2			5			10		
		n	25	50	100	25	50	100	25	50	100
distribution	alt										
Weibull	A1		6.3	5.7	5.7	6.0	5.9	5.1	5.5	5.5	5.1
	A2		6.0	6.1	5.8	6.7	5.3	4.5	6.0	5.5	5.5
	A3		7.9	7.9	6.9	7.8	6.2	5.5	6.6	5.5	6.3
normal	A1		5.8	5.2	5.5	5.8	5.6	5.1	5.1	5.8	5.0
	A2		5.7	6.0	5.6	5.9	5.8	4.8	4.4	5.8	5.1
	A3		8.8	8.4	7.0	7.0	6.4	5.8	5.1	5.8	5.6

Summary

- As expected, power increases with the sample size n.
- Perhaps surprisingly, the test has more power when m is small! Possibly due to alternative.
- Alternative 1 is often the most powerful (less degrees of freedom), despite misspecification.
- There is a small size distortion (up to about 5%) for alternative A_3 in small samples.

Stationary sequences with extremal clustering

To investigate serial dependence, we also consider data from first-order max-autoregressive (MAR) processes parametrized in terms of the extremal index θ.

Consider the stochastic process obtained through the recursion (Valadares Tavares 1977)

$$\begin{aligned} X_i &= \max\{X_{i-1}, Z_i\} + \log(1-\theta), \quad i \geq 1 \\ Z_i &\sim \mathsf{GEV}\{\log(\theta) - \log(1-\theta), 1, 0\} \end{aligned}$$

The margins are standard Gumbel, $X_i \sim \mathsf{GEV}(0, 1, 0)$, while the distribution of $Y = \max\{X_1, \ldots, X_m\}$ is Gumbel with location $\log[(m-1)\theta + 1]$.

Stationary setting

The extremal types theorem extends to stationary time series (Leadbetter 1974, 1983; Leadbetter et al. 1983).

Under a condition $D(u_n)$ and for extremal index $\theta \in (0, 1]$, we get instead

$$\lim_{m \to \infty} G^{\theta m}(c_m x + d_m) = F(x)$$

or equivalently

$$\lim_{m \to \infty} G^{m}(c_{\theta m} x + d_{\theta m}) = F(x).$$

Location parameters of stationary sequences

Max-stability holds asymptotically, but with $m\theta$ rather than m.

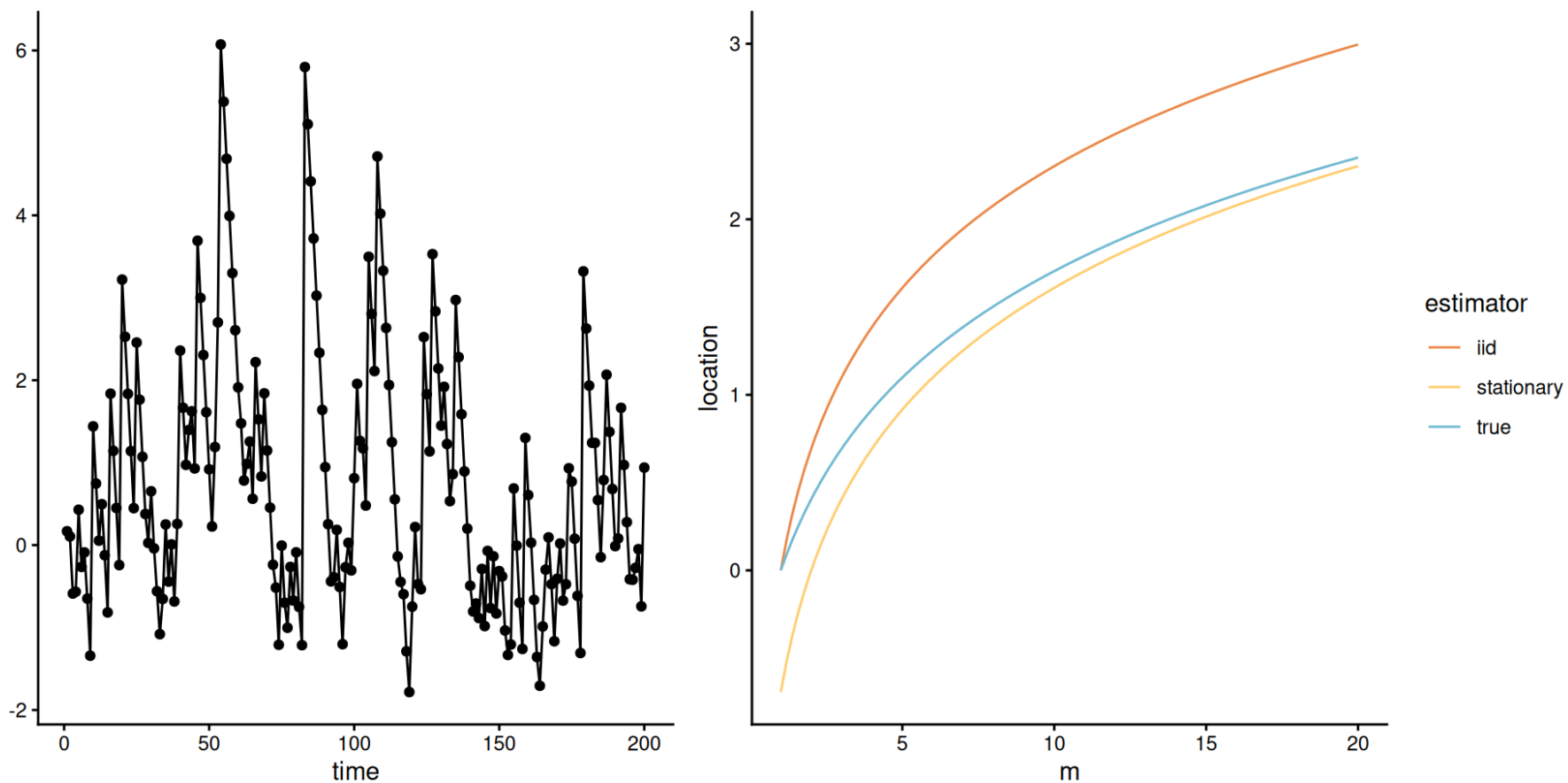
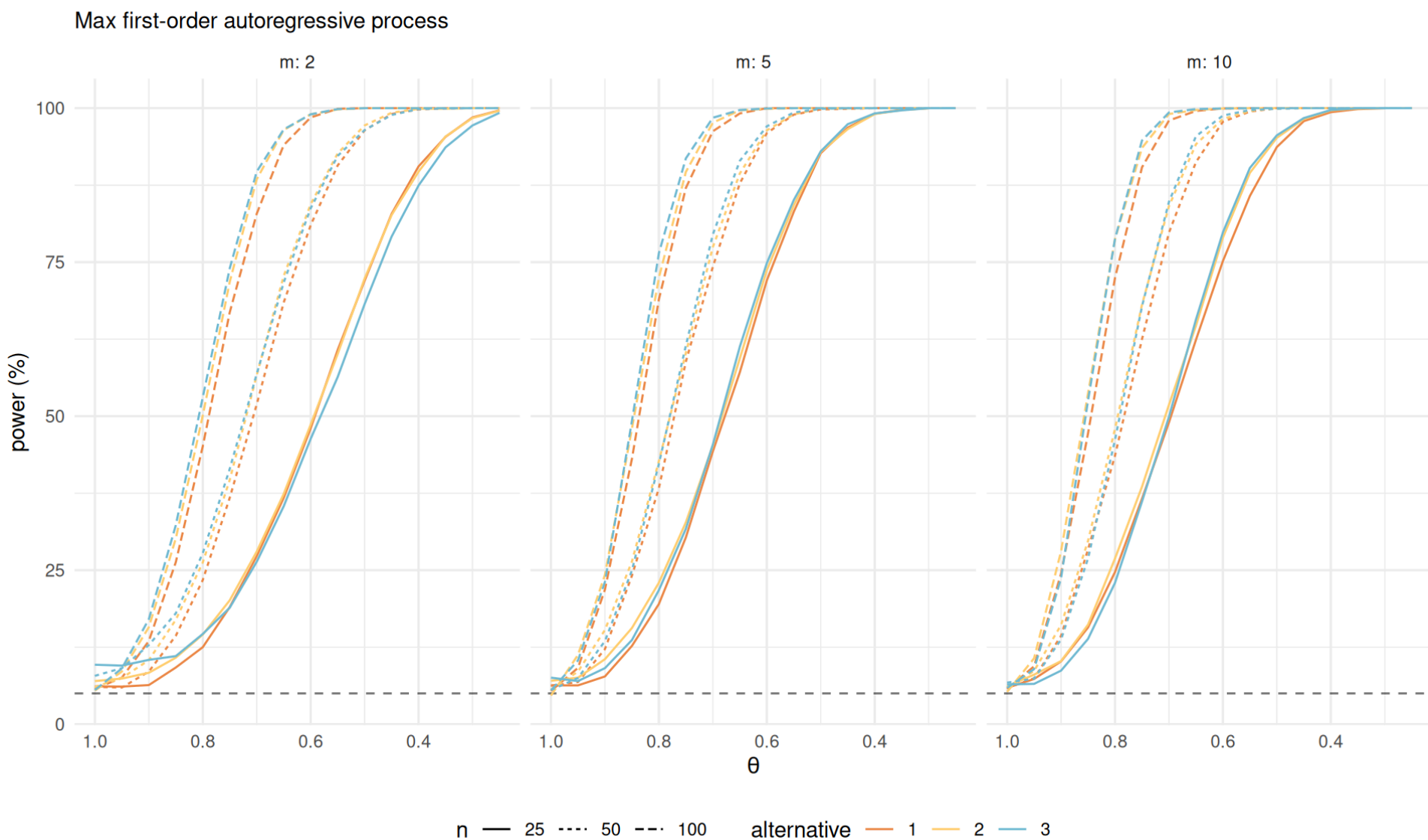
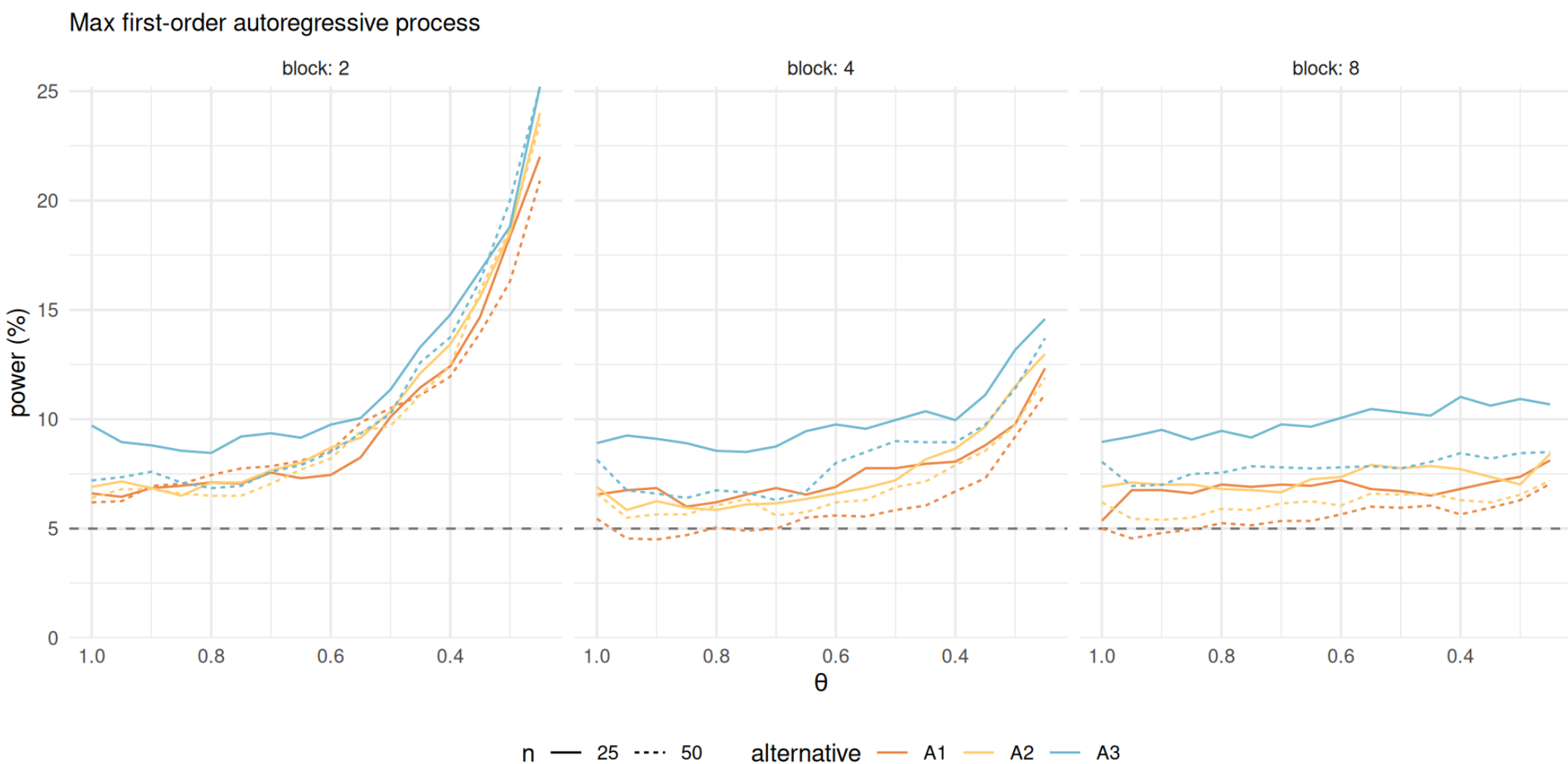


Figure 4: MAR with $\xi = 0$ and $\theta = 0.5$: time series of simulated values (left) and location parameters of the Gumbel maxima based on extrapolating to blocks of size m, taking $\log(m)$ (iid) and $\log(m\theta)$ (stationary) as location parameters.

Power curves - max autoregressive



Power study - max-autoregressive



Comments on power curves

- More discrepancies are expected for smaller values of θ, as the average cluster size is $1/\theta$, the reciprocal of the extremal index.
- As the block length increases, the effect of θ is captured through parameter estimates and we fail to reject for larger θ, as expected.
- Alternative 3 suffers again from size distortion, especially in small samples (either when m or n is small).

Diagnostic plots

We propose quantile-quantile plots to assess the goodness-of-fit of the model.

Among quantities of interest

- all maxima,
- the block maximum $X_{(m)}$.

We use a parametric bootstrap scheme for uncertainty quantification.

Confidence intervals for probability-probability plots

Säilynoja et al. (2022) outlines a method for building simultaneous confidence intervals for tests of uniformity in probability-probability plots, evaluated at N fixed plotting positions $\nu_1, \ldots, \nu_N \in (0, 1)$.

We propose a modification that accounts for estimation uncertainty and potentially unequal samples sizes due to rounding.

Building confidence intervals

- Consider the empirical distribution function (ECDF) $\mathsf{ECDF}(\nu; \boldsymbol{V}) = n^{-1} \sum_{i=1}^{n} \mathbf{1}_{\nu \leq V_i}$.
- If V_i are independent uniforms, then $n\mathsf{ECDF}(\nu; \boldsymbol{V})$ is binomial with n trials and probability of success ν, whose distribution we denote $H(\cdot; n, \nu)$.
- We can obtain a pointwise confidence interval at any value ν from the quantile function of the binomial.

Simultaneous intervals

To obtain simultaneous confidence interval for points $\nu_1, \ldots, \nu_N$, Säilynoja et al. (2022) consider simulation of B uniform samples of size n.

For the bth bootstrap, generate $\boldsymbol{v}_b \in [0,1]^n$ uniform and find the minimum level α_b such that the entire scaled ECDF lies inside the interval, where for $i = 1, \ldots, N$ and $b = 1, \ldots, B$

$$
\begin{aligned}
t_{l,i} &= H\left[n\mathsf{ECDF}(\nu_i; \boldsymbol{v}_b); n, \nu_i\right] \\
t_{u,i} &= 1 - H\left[n\mathsf{ECDF}(\nu_i; \boldsymbol{v}_b) - 1; n, \nu_i\right] \\
\alpha_b &= 2 \min_{i=1}^{N} \min\{t_{l,i}, t_{u,i}\}.
\end{aligned}
$$

For simultaneous intervals, use level α^*, the α quantile of $\alpha_1, \ldots, \alpha_B$.

Parametric bootstrap

A conventional parametric bootstrap to test properties of the statistic $S(\boldsymbol{\vartheta})$ based on the MLE $\widehat{\boldsymbol{\vartheta}}$ would

- simulate GEV via the quantile function, $X_j^* \sim \text{GEV}(\widehat{\mu}, \widehat{\sigma}, \hat{\xi})$.
- obtain bootstrap maximum likelihood estimates $\widehat{\boldsymbol{\vartheta}}^*$ from $\boldsymbol{X}^*$.
- Transform observations to uniform via the distribution function $F(X_j; \widehat{\boldsymbol{\vartheta}}^*)$ or $F[X_{(m)}; g(\widehat{\boldsymbol{\vartheta}}^*)]$.
- compute the ECDF of the resulting uniform sample and evaluate it at $\nu_1 = 1/N, \ldots, \nu_{N-1} = (N-1)/N$
- obtain α_b from the formula on the previous slide.

The procedure is repeated B times and we obtain the overall level $\alpha^* < \alpha$ for simultaneous confidence intervals.

Complications from rounding and censoring

For rounded and left-censored data, we modify the bootstrap scheme as follows:

- The n bootstrap draws are rounded to the nearest δ.
- The MLE is computed using likelihood with left-censoring and interval-censoring.
- We impute the latent variables X^* for the rounded values, and retain only those above u, leading to samples of size $n_b \leq n$.

Marginal likelihood

Since interest lies in extrapolation, we can also consider estimation of GEV parameters based on the marginal distribution of $X_{(1)}, \ldots, X_{(m-1)}$, where

$$\begin{aligned}\mathcal{L}_{\mathrm{marg}}(\boldsymbol{\theta}) &= \prod_{i=1}^{n} f_{(1),\ldots,(m-1)}(\boldsymbol{x}_i; \boldsymbol{\theta}) \\ &\propto \prod_{i=1}^{n} \{1 - F(x_{i,(m-1)})\} \prod_{j=1}^{m-1} f(x_{i,(j)}; \boldsymbol{\theta})\end{aligned}$$

which amounts to right-censoring the largest observation.

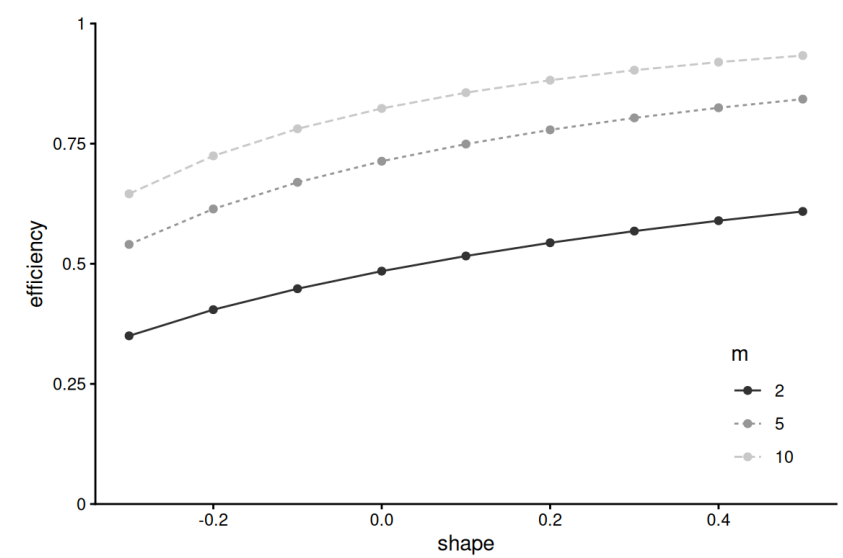
(Dis)advantages of the marginal likelihood

- Avoids using the maximum $X_{(m)}$ both for model estimation and goodness-of-fit: no double dipping.
- Need to add the inequality constraint $\sigma_0 + \xi_0(\max_i x_{i,(m)} - \mu_0) > 0$ on the optimization.

- Discarding the largest observation leads to a loss of information: with $m = 4$, we calculate the efficiency for $\boldsymbol{\vartheta} = (\mu = 0, \sigma = 1, \xi)$,

$$r(\boldsymbol{\vartheta}) = \frac{|\imath_{\text{marg}}(\boldsymbol{\vartheta})|^{1/3}}{|\imath_{\text{full}}(\boldsymbol{\vartheta})|^{1/3}},$$

where $\imath$ is the Fisher information.



Marginal likelihood

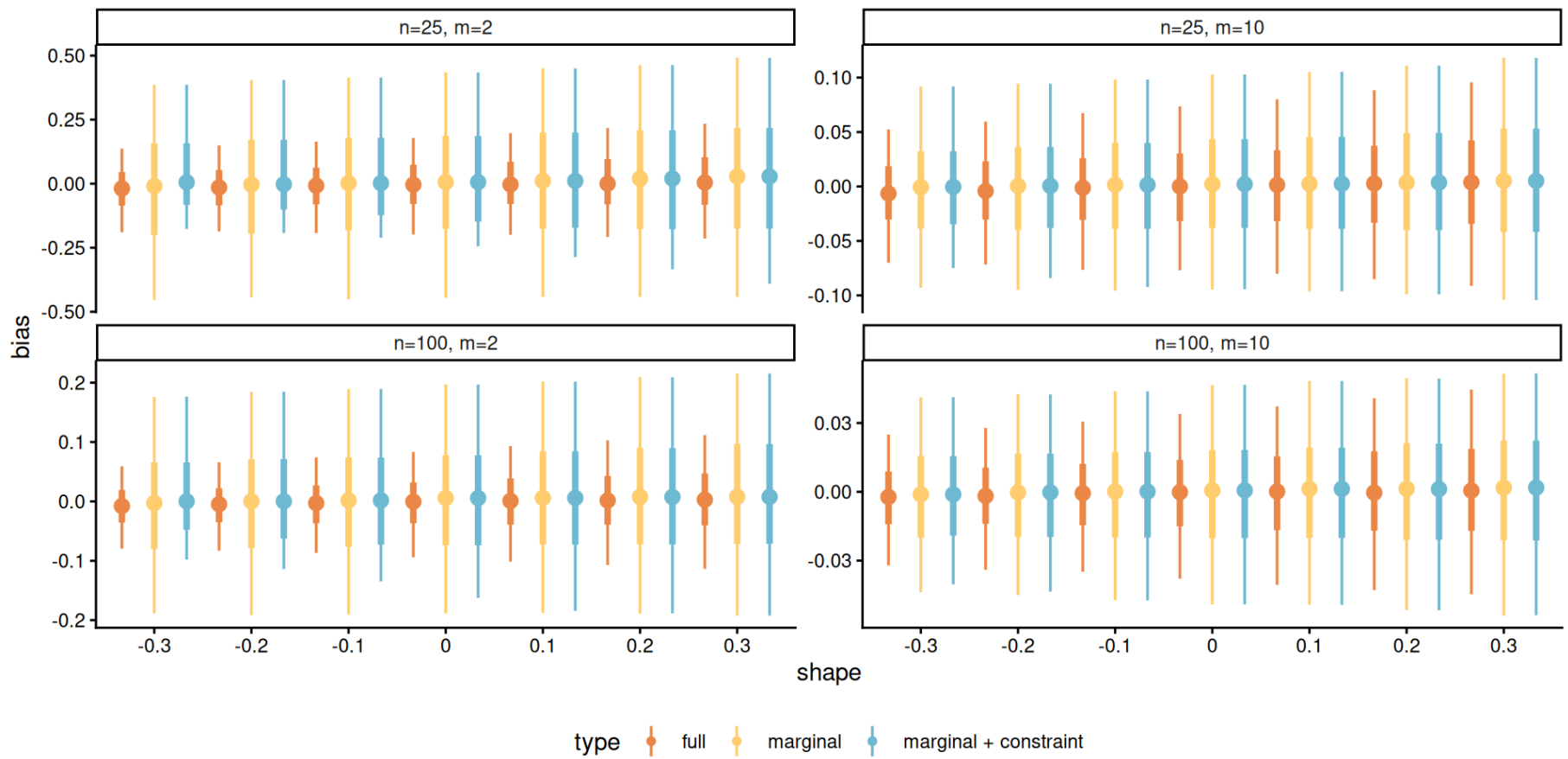


Figure 5: Interval plots showing median, 50% and 90% intervals of the bias of maximum likelihood estimators of the shape parameters for datasets drawn from a GEV with shape ξ for $m \in \{2, 10\}$ and $n \in \{25, 100\}$.

Data applications

Cheeseboro wind gusts

We use daily data from the Western Regional Climate Center for the Cheeseboro weather station.

The data consist of daily maximum gust wind speed (in meters per second) for 1 January to 1 February inclusive between 1996 and 2026, a total of 961 records.

We test for different block sizes $m \in \{1, 2, 4, 8, 16, 32\}$ days.

Diagnostic plots

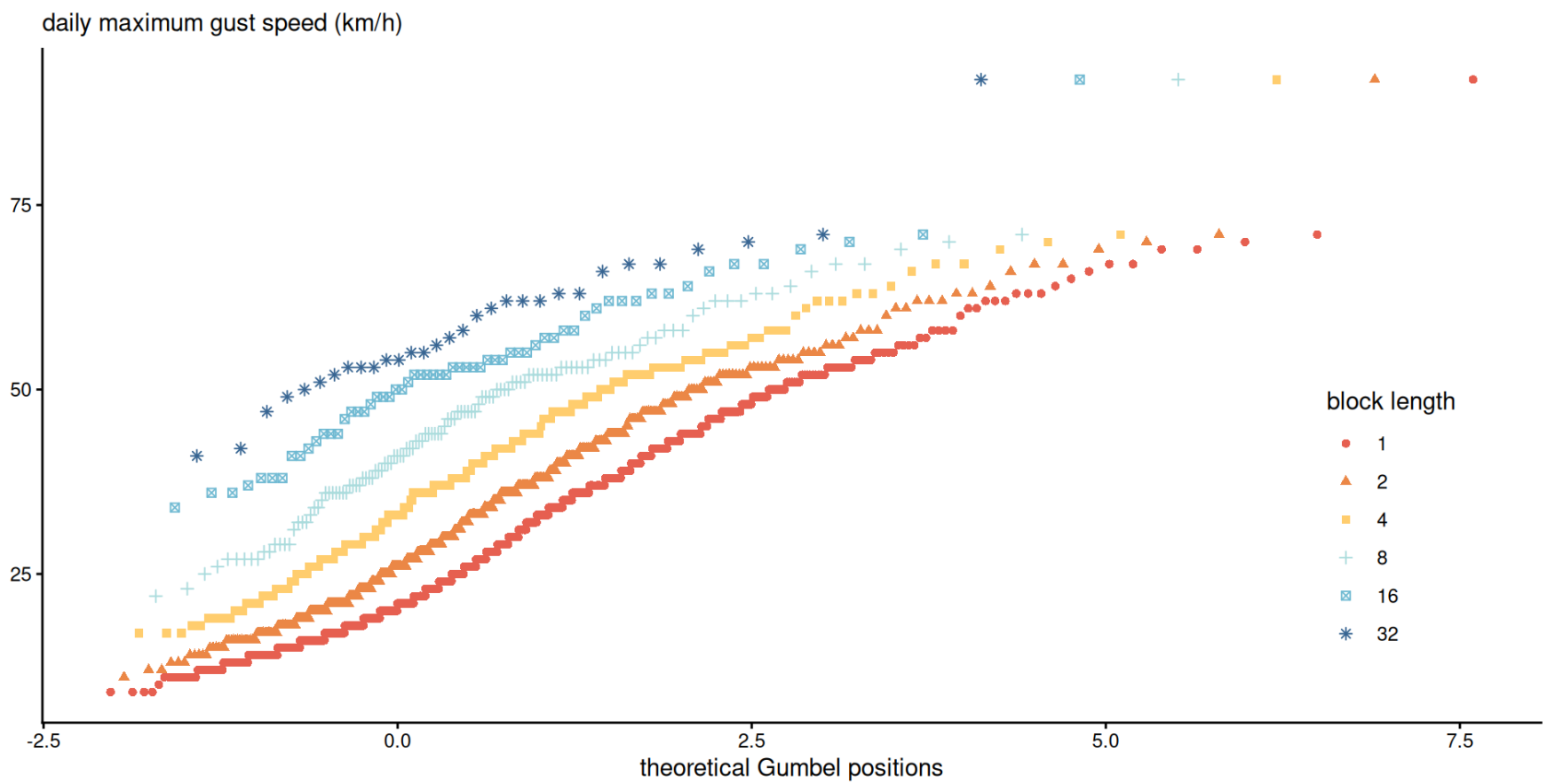


Figure 6: Gumbel plot of m-day maxima against standard Gumbel plotting positions $-\log[-\log\{i/(n_m+1)\}]$ for the Cheeseboro wind gust data.

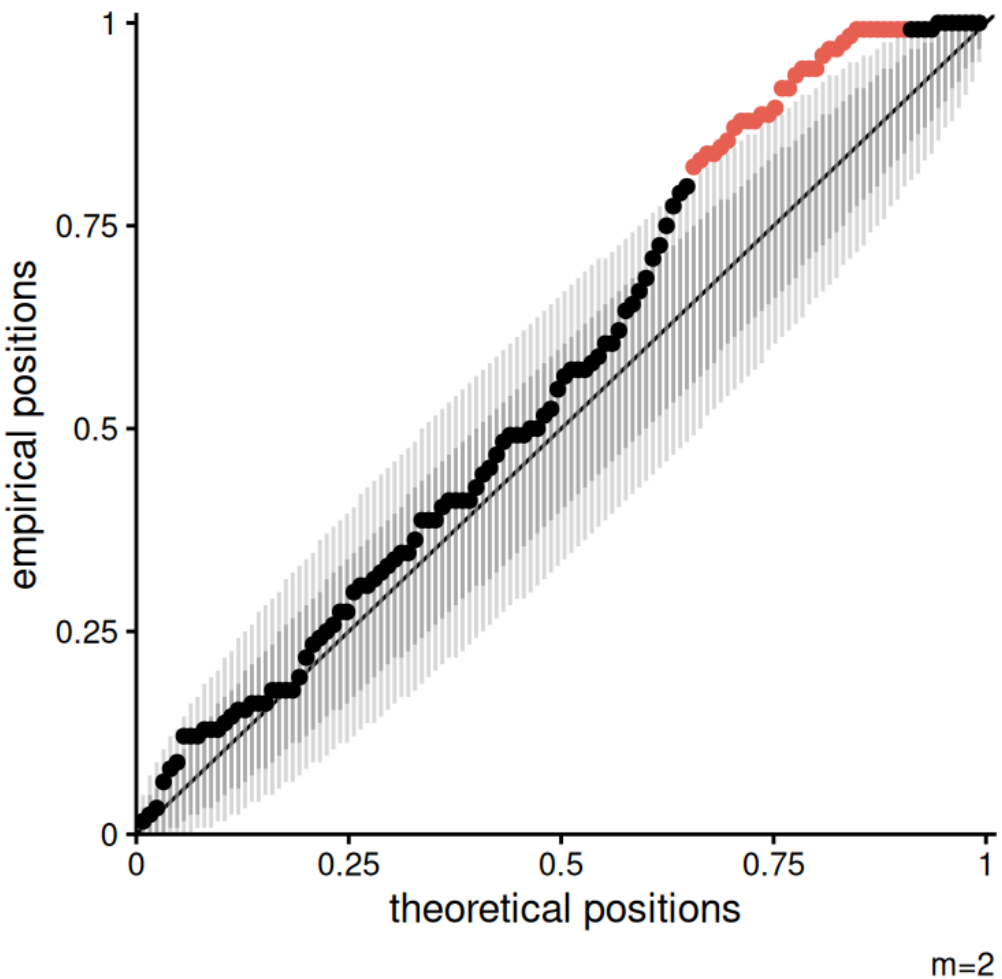
Plot *à la* Cox et al. (2002). With Gumbel fit, lines should be parallel.

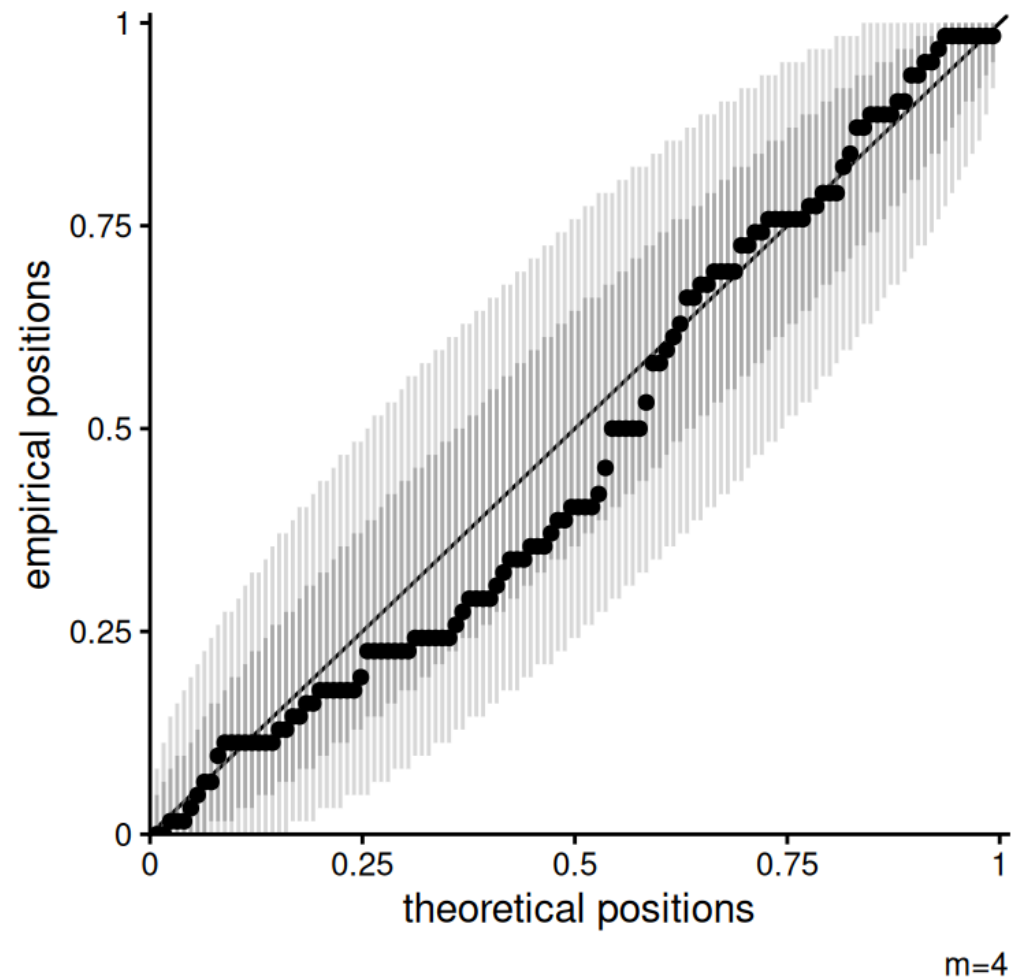
P-values for test for wind gusts at Cheeseboro

Table 2: P-values (rounded to three digits) for the test of max-stability.

	$c = 2$			$c = 4$		
block	A_1	A_2	A_3	A_1	A_2	A_3
$m = 1$	0	0	0	0	0	0
$m = 2$	0.013	0.045	0	0.12	0.29	0
$m = 4$	0.373	0.537	0.16	0.19	0.4	0.18
$m = 8$	0.135	0.316	0.45	0.64	0.76	0.81

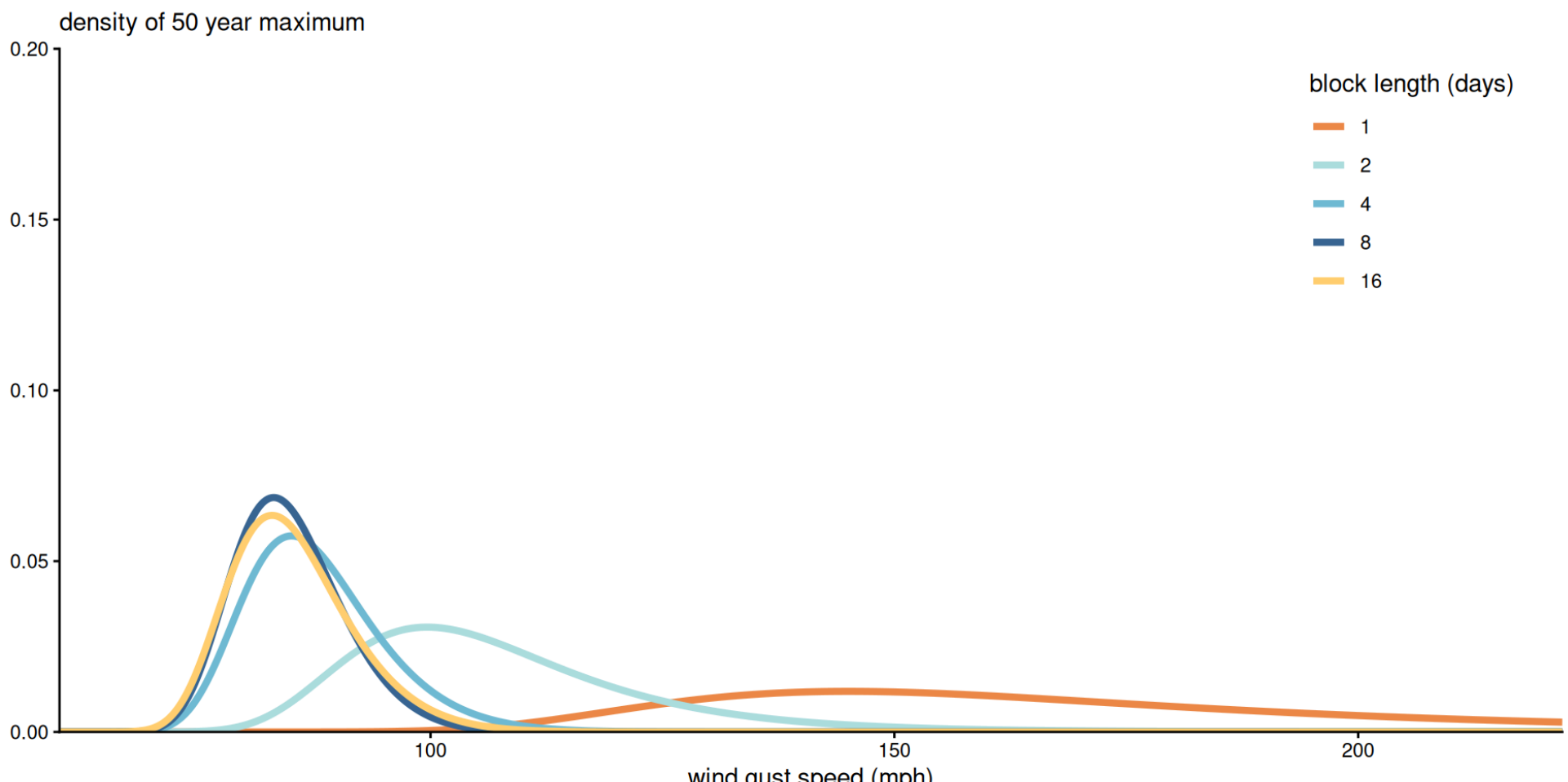
Probability-probability plots of maxima





Risk extrapolation

The block length can have a sizeable impact on risk estimation (sample maximum is 92 mph $\sim$ 148km/h). Shape of $\hat{\xi} = 0.18$ for $m = 1$, versus $\xi = -0.09$ for $m = 4$.



Inference for extremes

Use GEV model fitted to blocks of size 4 days to calculate risk measures for the maximum 50 years wind gust, by using the max-stability relationship with $m = 8 \times 50$.

- The estimated median of half-century maximum daily wind gust is 86.73 miles per hour
- 95% profile-based confidence intervals: (83.06, 105.07) mph.
- Negligible serial correlation.

Application 2: landslide-triggering rainfall episodes

We look at data from the Abisko scientific research station, located in Lapland (Sweden).

- The daily data run from January 1913 until December 2014.
- The region around the station is prone to landslides, which are caused by prolonged heavy rainfall episodes.
- Guzzetti et al. (2007) identify intensity-duration combinations that are likely to lead to landslides, for three day rainfall, rainfall episodes beyond 69.9mm;

We compute 3 day running sums and consider rainfall with 12 periods of 7 days from June 15th every year to ensure approximate stationarity.

Tests for block-length

Since rainfall can be small, we left-censor maxima below 10 mm.

- The proportion of points being left-censored is 0.62 for weekly maxima, 0.19 for monthly and none for seasonal.

P-values for the test of max-stability: comparing

- 7-day to 28-day maxima ($m = 4$) yields p-values less than 10^{-15}.
- 28-day to seasonal ($m = 3$) leads to p-values of 0.41 for A_1, 0.41 for A_2 and 0.79 for A_3.

Risk assessment for Abisko rainfall

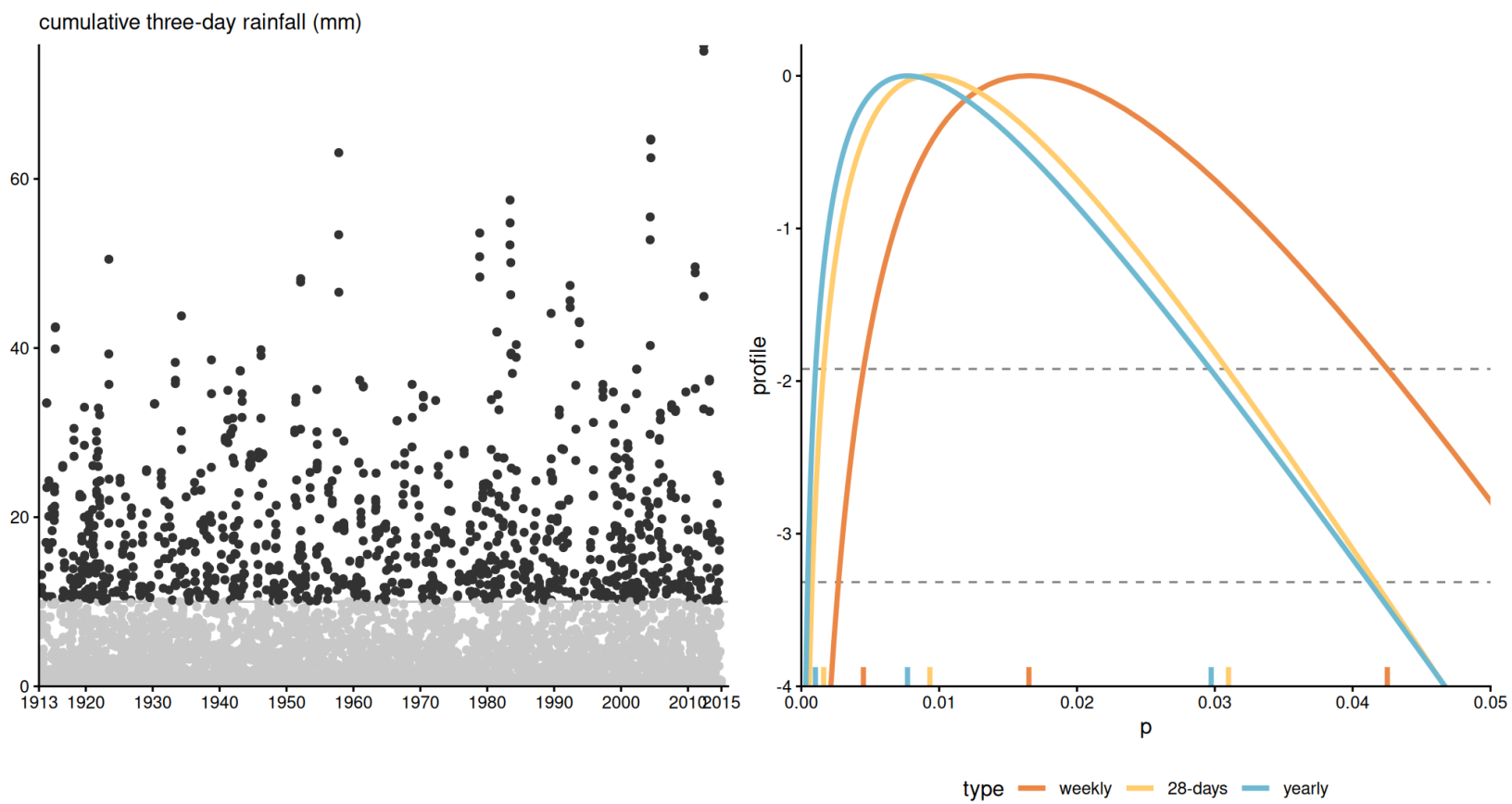


Figure 7: Time series of three-day cumulated rainfall measurements (in mm) and profile likelihood for the probability of having a summer rainfall exceeding 69.9mm based on the GEV model fitted to weekly, 28-days and yearly maxima (right).

Summary

We propose likelihood ratio tests for testing max-stability for blocks of size m

- Build block of $n/m \times m$ ordered data from maxima.
- Check fit to largest (max of m) and use max-stability to detect departures.
- Simulation studies show the test is powerful even in small samples.
- The test with a single parameter seems more powerful than competitors, with lower values of m.

Diagnostic plots based on bootstrap to account for goodness-of-fit.

Limitations

- We consider only block maxima of length $m_2 = cm_1$ for $m_1, m_2, c \in \mathbb{N}$ (multiples):
  - does not allow for comparison of blocks of size 2 vs 3, say.
  - some data discarded if n is not divisible by m (but could use ideas from Simpson and Northrop (2026)).
  - in practice it is natural to consider aggregation of periods, e.g., monthly to yearly.
- The scheme leads to multiple testing if multiple values of m and block lengths are considered:
  - smaller values of m should lead to higher power, as shown through simulations,
  - but the choice of m should be dictated more by practical considerations.
- Our approach is restricted to independent or stationary time series.

Extensions and future work

- Combining model with estimates of the extremal index θ.
- Alternative estimators for the GEV built from sliding blocks (e.g., Bücher and Zanger 2023).

Theory for these is more complicated, owing to lack of genuine likelihood.

- Could consider Wald-based tests with sandwich-based or bootstrap covariance estimators.

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